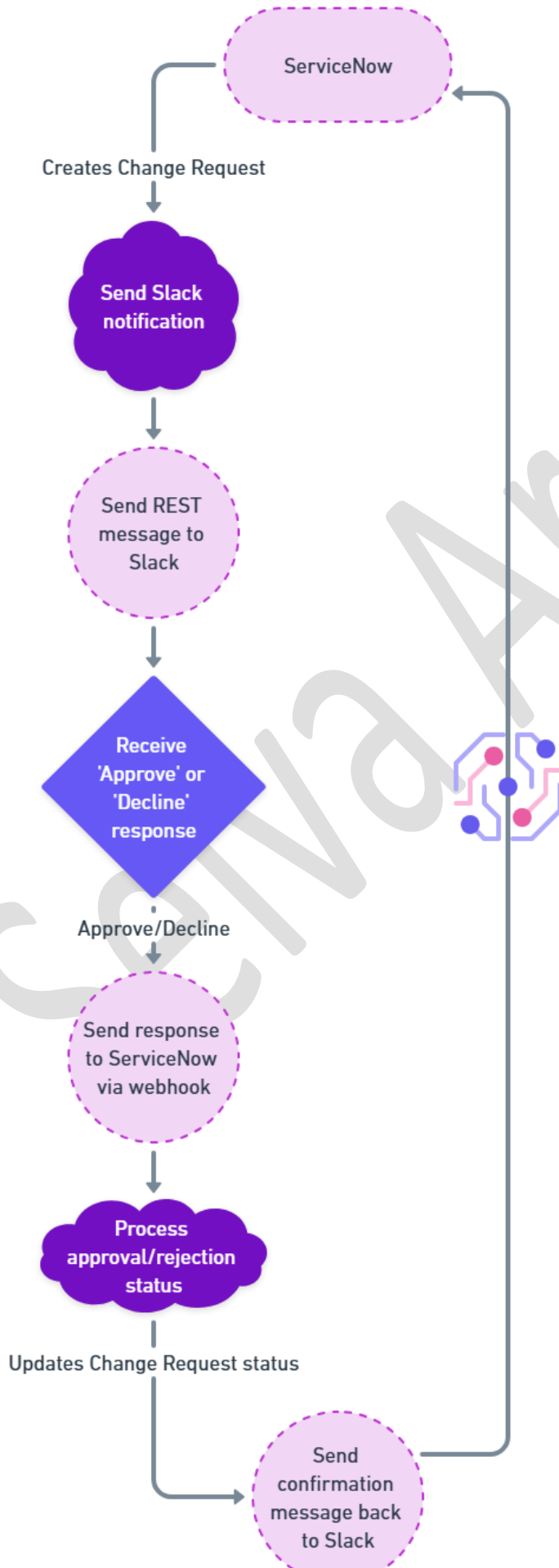


Custom ServiceNow to Slack Integration for Change Request Approvals

This guide will walk you through setting up a custom integration between ServiceNow and Slack. The integration will:

-  **Send a Slack message** when a Normal Change Request is created in ServiceNow.
-  Allow approvers to **Approve/Decline directly in Slack**.
-  Automatically **update the Change Request status** in ServiceNow.
-  **Post a response back to Slack**, stating that the Change Request has been updated after approval or rejection.



◆ Create a Slack App

We need to create a custom Slack App that sends and receives messages.

Step 1: Create a Slack Bot

1. **Go to Slack API Apps:** Visit the [Slack API Apps](#) page.
2. **Click "Create New App" → Select "From Scratch".**
3. **Enter App Name:** For example, ServiceNow Approvals.
4. **Select your Slack Workspace:** Choose the workspace where the bot will be installed.

Step 2: Enable Slack API Features

1. **Go to OAuth & Permissions.**
2. **Add the following Bot Token Scopes:**
 - ☒ chat:write (Send messages to channels)
 - ☒ commands (Create interactive buttons)
 - ☒ incoming-webhook (Receive responses)

Scopes

A Slack app's capabilities and permissions are governed by the [scopes](#) it requests.

Bot Token Scopes

Scopes that govern what your app can access.

OAuth Scope	Description	
channels:read	View basic information about public channels in a workspace	
chat:write	Send messages as @ServiceNow	
chat:write.public	Send messages to channels @ServiceNow isn't a member of	
groups:read	View basic information about private channels that ServiceNow has been added to	
im:write	Start direct messages with people	
incoming-webhook	Post messages to specific channels in Slack	

Add an OAuth Scope

User Token Scopes

Scopes that access user data and act on behalf of users that authorize them.

OAuth Scope	Description	
admin	Administer a workspace	
channels:read	View basic information about public channels in a workspace	
chat:write	Send messages on a user's behalf	
im:write	Start direct messages with people on a user's behalf	

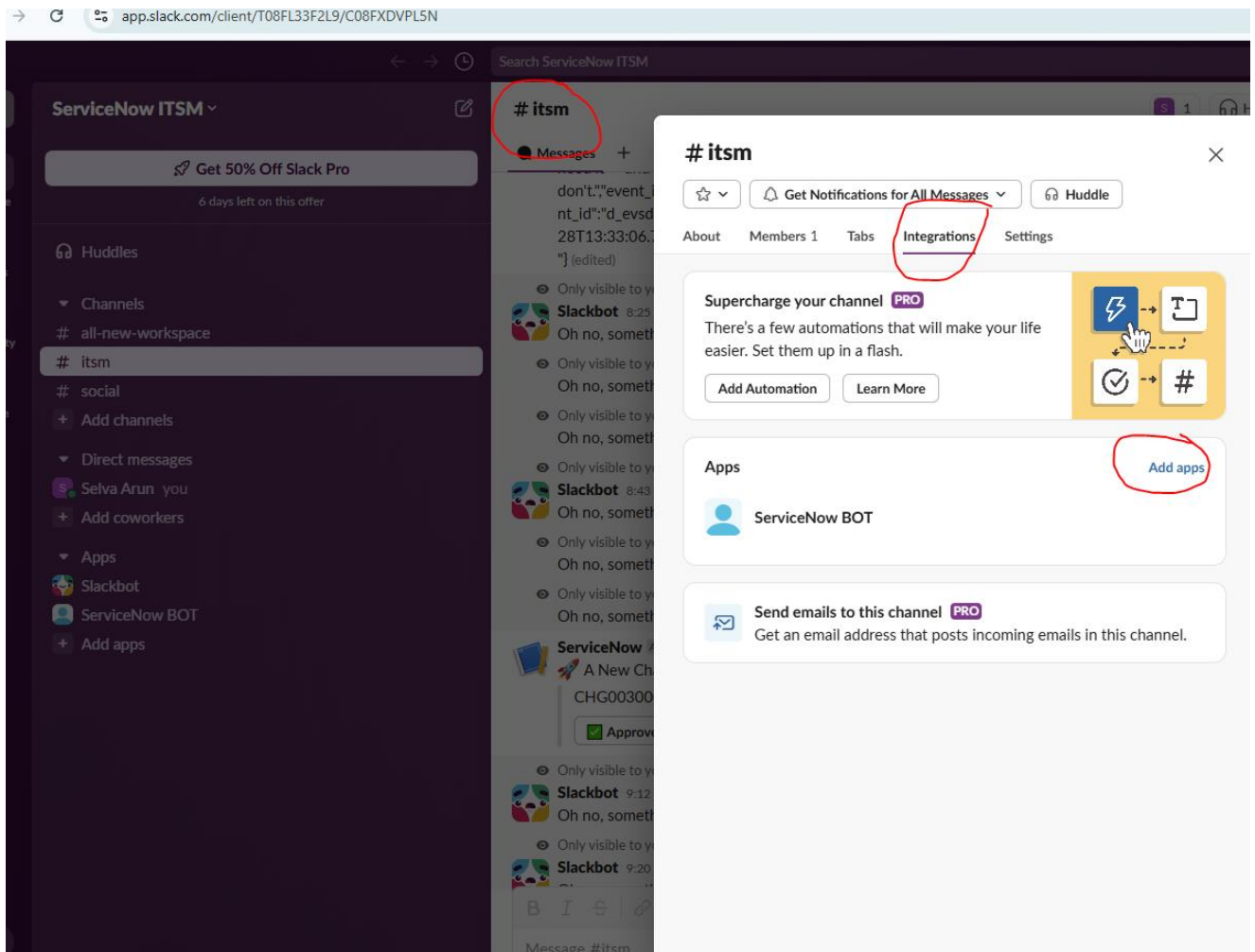
Add an OAuth Scope

3. Install the App to Your Workspace:

- Click **Install App**.
- Copy the **Bot Token** (starts with xoxb-...).

Step 3: Invite the App to the Channel

1. Invite the newly created Slack bot to the channel (in this case, the #itsm channel) Right click on the channel and click on Integration and add apps.



◆ 2 Configure ServiceNow to Send Slack Messages

We will use **ServiceNow Flow Designer** and **REST Messages** to send Slack notifications when a Change Request is created.

Step 1: Create a REST Message to Send Messages to Slack

1. **Navigate to:** System Web Services → Outbound → REST Message.
2. Click **New** and enter:
 - **Name:** Slack - Send Message
 - **Endpoint:** <https://slack.com/api/chat.postMessage>
 - **HTTP Method:** POST
3. **Add HTTP Headers:**

- **Authorization:** Bearer <YOUR_SLACK_BOT_TOKEN>
- **Content-Type:** application/json

HTTP Method
Slack - Send Message

REST Message Slack - Send Message (OAuth) ⓘ

* Name Slack - Send Message

* HTTP method POST ▼

Endpoint https://hooks.slack.com/services/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

Authentication HTTP Request

Use MID Server 🔍

HTTP Headers

Name ▲	Value
✕ Authorization	Bearerxoxb-8530105512689-8503865833143-t...
✕ Content-Type	application/json
+ Insert a new row...	

4. **Set the Request Body (JSON):**

```

5. {
6.   "channel": "#itsm",
7.   "text": "🚀 A New Change Request Needs Approval!",
8.   "attachments": [
9.     {
10.      "text": "${changeRequestId} - ${changeRequestDescription}",
11.      "fallback": "Approve or Reject",
12.      "callback_id": "${changeRequest}",
13.      "actions": [
14.        {
15.          "name": "approve",
16.          "text": "✅ Approve",
17.          "type": "button",
18.          "value": "approved"

```

19. },
20. {
21. "name": "decline",
22. "text": "❌ Decline",
23. "type": "button",
24. "value": "declined"
25. }
26.]
27. }
28.]
29. **Save & Test:** Ensure the message appears in your Slack channel.

HTTP Method
Slack - Send Message

```
{
  "text": "${changeRequestId} - ${changeRequestDescription}",
  "fallback": "Approve or Reject",
  "callback_id": "${changeRequest}",
  "actions": [
    {
      "name": "approve",
      "text": "✅ Approve",
      "type": "button",
      "value": "approved"
    },
    {
      "name": "decline",
      "text": "❌ Decline",
      "type": "button",
      "value": "declined"
    }
  ]
}
```

UpdateDelete

Related Links

[Auto-generate variables](#)
[Preview Script Usage](#)
[Set HTTP Log Level](#)
[Test](#)
[\[SN Utils\] Versions \(6\)](#)

Variable Substitutions (3)Test Runs (10)

≡ ▾

Name ▾

Search

⚙

Method = Slack - Send Message

☐ 🔍

Name ▲

Escape type

Test value

changeRequest

No escaping

CHG003

changeRequestDescription

No escaping

hello

changeRequestId

No escaping

CHG003

Test Runs
Created 2025-02-27 18:53:40

* Name

Slack - Send Message

HTTP status

200

Endpoint

https://hooks.slack.com/services/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

Parameters

Content

```
{
  "channel": "#itsm",
  "text": "🔔 A New Change Request Needs Approval!",
  "attachments": [
    {
      "text": "CHG0030001 - hello",
      "fallback": "Approve or Reject",
      "callback_id": "CHG0030001",
      "actions": [
        {
          "name": "approve",
          "text": "✅ Approve",
          "type": "button",
          "value": "approved"
        },
        {
          "name": "decline",
          "text": "❌ Decline",
          "type": "button",
          "value": "declined"
        }
      ]
    }
  ]
}
```

Response

ok

Update

Delete

◆ 8 Capture Approvals in Slack and Update ServiceNow

Slack buttons require a webhook endpoint to send responses back to ServiceNow, which will process the approval or rejection and update the Change Request status.

Step 1: Create an Inbound Webhook in ServiceNow

1. Go to: **System Web Services** → **Scripted REST APIs**.
2. Click **New**:
 - **Name**: Slack Change Approval
3. Check **Override default supported request formats** and **Override default supported response formats** to:
 - `application/x-www-form-urlencoded`

* Name

* API ID

Active ☒

Protection policy -- None --

Application Global

API namespace 1052697

Base API path /api/1052697/slack_change_approval

Security Content Negotiation Documentation

Default ACLs may be selected to apply to all resources, but individual resources can override this setting.

The Default ACLs are enforced for a resource when:

- The resource 'Requires authentication' and 'Requires ACL authorization' fields are selected, and
- The resource itself does not reference any ACL records

Access is granted if at least one matching ACL record is found.

[More info](#)

Default ACLs  Scripted REST External Default

[Update](#) [Delete](#)

Related Links

[Enable versioning](#)

[Explore REST API](#)

[API analytics](#)

[\[SN Utils\] Versions \(2\)](#)

Resources (1) Request Headers (1) Query Parameters

Search

API definition = Slack Change Approval

☐	Name	HTTP method	Relative path	Resource path
☐	process_approval	POST	/process_approval	/api/1052697/slack_change_approval/proce...

1 to 1 of 1

* Name

* API ID

Active ☒

Protection policy -- None --

Application Global

API namespace 1052697

Base API path /api/1052697/slack_change_approval

Security Content Negotiation Documentation

By default, all resources in an API support the following request and response formats: application/json, application/xml, and text/xml.

The default supported formats can be overridden here at the API level and will apply to all resources belonging to the API, unless an individual resource overrides the defaults.

[More info](#)

Override default supported request formats ☒

* Default supported request formats

Override default supported response formats ☒

* Default supported response formats

[Update](#) [Delete](#)

[Enable versioning](#)

[Explore REST API](#)

[API analytics](#)

[\[SN Utils\] Versions \(2\)](#)

Resources (1) Request Headers (1) Query Parameters

Search

API definition = Slack Change Approval

☐	Name	HTTP method	Relative path	Resource path	API version	Active
☐	process_approval	POST	/process_approval	/api/1052697/slack_change_approval/proce...	(empty)	true

1 to 1 of 1

4. Create a new Resource (POST method):

- **Name:** process_approval
- **Relative Path:** process_approval

Scripted REST Resource
process_approval

API definition Slack Change Approval ⓘ

* Name process_approval

Application Global

Active ☒

Request routing

The route configuration specifies the 'HTTP method' and 'Relative path'. These fields determine how HTTP clients access this resource.

The relative path identifies the sub-path to this resource relative to the base API path. The relative URI can contain path parameters such as '/abc/{id}'. The requesting client specifies the id value, available to the script at runtime.

[More info](#)

* HTTP method POST

Relative path /process_approval

Resource path /api/1052697/slack_change_approval/process_approval

Implement the resource

Access request details including URI path parameters, query parameters, headers, and the request body using the: [Request API](#).

Configure the response including setting the HTTP status code, response body, and any response headers using the: [Response API](#).

[More info](#)

* Script Turn on ECMAScript 2021 (ES12) mode ⓘ

```

5
6     if (!payloadString) {
7         gs.error("Missing payload parameter.");
8         response.setStatus(400);
9         response.setHeader("Content-Type", "application/x-www-form-urlencoded");
10        response.setBody("error=" + encodeURIComponent("Missing payload parameter"));
11        return;
12    }
13
14    // Attempt to parse the payload JSON
15    var payloadObj;
16    try {
17        payloadObj = JSON.parse(payloadString);
18    } catch (e) {
19        gs.error("Error parsing queryParams payload JSON: " + e);
20        response.setStatus(400);
21        response.setHeader("Content-Type", "application/x-www-form-urlencoded");
22        response.setBody("error=" + encodeURIComponent("Invalid JSON format in payload"));
23        return;
24    }
25
26    // Extract approval details
27    var approvalStatus = payloadObj.actions[0].value;

```

Step 2: Insert the Script to Process Slack Responses

Add the following script to handle incoming responses from Slack and update the Change Request status in ServiceNow:

```

(function process(request, response) {
    try {
        var payloadString = request.queryParams.payload;

        if (!payloadString) {
            gs.error("Missing payload parameter.");

```

```
response.setStatus(400);  
response.setHeader("Content-Type", "application/x-www-form-urlencoded");  
response.setBody("error=" + encodeURIComponent("Missing payload parameter"));  
return;  
}
```

```
var payloadObj;  
try {  
    payloadObj = JSON.parse(payloadString);  
} catch (e) {  
    gs.error("Error parsing payload: " + e);  
    response.setStatus(400);  
    response.setHeader("Content-Type", "application/x-www-form-urlencoded");  
    response.setBody("error=" + encodeURIComponent("Invalid JSON format in payload"));  
    return;  
}
```

```
var approvalStatus = payloadObj.actions[0].value;  
var changeRequestNumber = payloadObj.callback_id;  
  
var gr = new GlideRecord('sysapproval_approver');  
gr.addQuery('sysapproval.number', changeRequestNumber);  
gr.query();
```

```
if (gr.next()) {  
    gr.state = approvalStatus === "approved" ? 'approved' : 'rejected';  
    if (approvalStatus === "declined") gr.comments = "Declined via Slack";  
    gr.update();  
    gs.info("Change Request " + changeRequestNumber + " updated successfully.");  
}
```

```

var slackMessage = {
    "channel": "#itsm", // Specify the channel to post the message in
    "text": "Change request " + changeRequestNumber + " has been " + approvalStatus + " as
per your response.",
    "response_type": "in_channel" // Ensures the message is visible in the channel
};

var r = new sn_ws.RESTMessageV2('Slack - Send Message (OAuth)', 'Slack - Send Message');
r.setRequestBody(JSON.stringify(slackMessage));
var res = r.execute();
var responseBody = res.getBody();
var statusCode = res.getStatusCode();
gs.info("Slack response: " + responseBody + " Status Code: " + statusCode);

response.setStatus(200);
response.setHeader("Content-Type", "application/x-www-form-urlencoded");
response.setBody("success=true&message=" + encodeURIComponent("Approval status
updated"));
} else {
    gs.error("Change Request not found: " + changeRequestNumber);
    response.setStatus(404);
    response.setHeader("Content-Type", "application/x-www-form-urlencoded");
    response.setBody("error=" + encodeURIComponent("Change request not found"));
}
} catch (e) {
    gs.error("Error processing Slack approval: " + e.toString());
    response.setStatus(500);
    response.setHeader("Content-Type", "application/x-www-form-urlencoded");

```

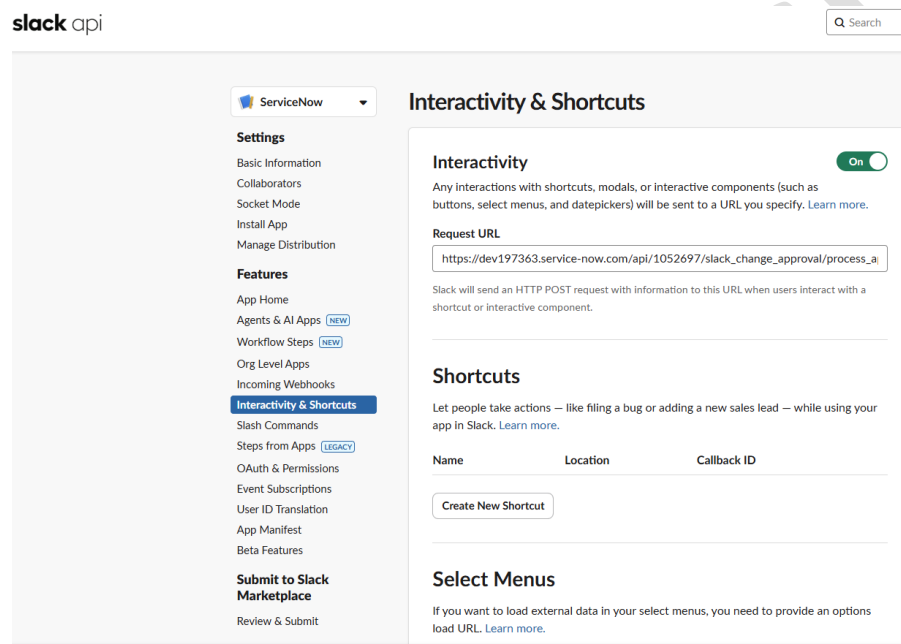
```

        response.setBody("error=" + encodeURIComponent("Internal server error"));
    }
})(request, response);

```

Step 3: Connect Slack to ServiceNow Webhook

1. **Go to Slack App Settings → Interactivity & Shortcuts.**
2. **Enable Interactivity:**
 - **Request URL:**
`https://YOUR_SERVICENOW_INSTANCE/api/x_snc_slack/change_approval/process_approval`
3. **Save Changes.**



◆ 3 Automating the Process with Flow Designer and Integration Hub:

Step 1: Activate Integration Hub Plugins

1. **Navigate to:** System Definition → Plugins.
2. **Search for the following plugins:** ServiceNow IntegrationHub Starter Pack Installer, ServiceNow IntegrationHub Standard Pack Installer /ServiceNow IntegrationHub Professional Pack Installer / ServiceNow IntegrationHub Enterprise Pack Installer (install any one of these).
3. **Click Activate** next to each of these plugins.

Step 2: Create an Action in Flow Designer

Once Integration Hub is activated, you need to create an action that will be used to send the Slack message.

1. Go to Flow Designer:

- **Navigate to:** Flow Designer → Designer.
- **Click New** to create a new action.

2. Create the Action:

- **Name:** Send Slack Approval Request
- **Table:** Select Global or a specific table that aligns with your Change Request.
- **Type:** Select **Action**.

3. Add a Step to Call the REST Message:

- **Click the + sign** to add a new step.
- **Choose the REST Message option** and create a credential by clicking on the + icon next to credential alias, select Basic Auth and Give your slack username and BOT Token and select the connection as Define Connection Inline.

1. REST step

REST {REST} Data Collapse All

Input Variables

changeRequestId

changeRequest

changeRequestDesc

REST step

Open Record

Connection Details

Connection Define Connection Inline

Use MID

Credential Alias Slack_Token

Base URL https://hooks.slack.com/services/T08FL33F2L9/B08FL7SMHMF/t7s2OxHrTDhxWOXcsqGPnE9

Connection Timeout

Request Details

Build Request From REST Message

REST Message Slack - Send Message (OAuth)

REST Message Function Slack - Send Message

Resource Path

HTTP Method POST

Query Parameters

Name

Value

Connection & Credential Aliases

* Name Slack Token

ID Slack_Token

Application Global

Parent Alias

* Type Credential

Connection type HTTP

Support Multiple Active Connections

Default Retry Policy

Configuration Template

- Select the previously created REST message (Slack - Send Message).
- Map the action inputs to the REST message fields:
 - Channel: Use the Channel input.
 - Text: Populate the text dynamically with the necessary information (like Change Request number).

Slack Post API

scriptsync

1. REST step

REST {REST}

Inputs

REST step

Error Evaluation

Outputs

Connection Details

ConnectionDefine Connection Inline

Use MID

Credential AliasSlack_Token

Base URLhttps://hooks.slack.com/services/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

Connection Timeout

Test REST Step

Request Details

Build RequestFrom REST Message

REST MessageSlack - Send Message (OAuth)

REST Message FunctionSlack - Send Message

Resource Path

HTTP MethodPOST

Query Parameters

If this step failsStop the action and go to error evaluation

Slack Post API

1. REST step

Inputs

1 (REST) REST step

Error Evaluation

Outputs

Headers

AuthorizationBearerxoxb-8530105512689-850386tnSceAnXlFON7U2Wm2mctl0n

Content-Typeapplication/json

Request Content

Request TypeText

Request Body[Text]

```
{
  "name": "approve",
  "text": "✅ Approve",
  "type": "button",
  "value": "approved"
},
{
  "name": "decline",
  "text": "❌ Decline",
  "type": "button",
  "value": "declined"
}
]
```

9413 characters remaining

If this step failsStop the action and go to error evaluation

4. The data from the REST message gets copied over to the flow Rest Step which would automatically create the variables too. Save the Action and Publish it.

Step 3: Use the Action in a Flow

Now that the action has been created, it needs to be used within a flow to automate the Slack notification process.

1. **Create a New Flow:**
 - **Navigate to:** Flow Designer → Designer.
 - **Click New** to create a new flow.
2. **Define the Flow Trigger:** Choose a trigger that will initiate the flow. For a Change Request, you can use the **Record Triggered Flow**.
 - **Table:** Change Request.
 - **Trigger:** Select Created or any relevant trigger (e.g., when the Change Request enters a specific state like "Approval").

 **Notify Slack on Change Request Authorization** Active

TRIGGER



Change Request Updated where (State is Assess, and Assignment group is ITSM App-Dev)

Trigger

Updated

* Table

Change Request [change_request] X

Condition

All of these conditions must be met

AND

State

is

Assess

Assignment group

is

ITSM App-Dev

or

New Criteria

Run Trigger

Once

Advanced Options

3. **Add the Action:**
 - In the **Flow Steps**, click the + sign and select **Action**.
 - Choose the action you created earlier (Send Slack Approval Request).

- Map the necessary fields from the Change Request record to the inputs for the action (e.g., Channel, Change Request Number, etc.).

ACTIONS [Select multiple](#)

1  Slack Post API 

Action Properties

Action

Slack Token Connection  

Action Inputs

changeRequestId  

changeRequest  

changeRequestDescription  

 [Add an Action, Flow Logic, or Subflow](#)

4. Save and Activate the Flow.

◆ 4. Testing the Integration

1. Create a Normal Change Request in ServiceNow.

Change Request
CHG0030011

New ✓ Assess Authorize Scheduled Implement Review Closed

Number: CHG0030011

Requested by: System Administrator

Category: Other

Service:

Service offering:

Configuration item:

Priority: 4 - Low

Risk: Moderate

Impact: 3 - Low

Short description: Apply the security patch to the servers

Description:

Model: Normal

Type: Normal

State: Assess

Change is waiting for approval

On hold: ☐

Conflict status: Not Run

Conflict last run:

* Assignment group: ITSM App-Dev

Assigned to:

Planning Schedule Conflicts Notes Closure Information

Planned start date and Planned end date are the requested change window

Planned start date: 2025-03-04 11:32:42

Planned end date: 2025-03-05 11:32:46

CAB required: ☐

CAB date/time:

Actual start date:

Actual end date:

CAB delegate:

CAB recommendation:

2. Slack will send an approval message to the #itsm channel.

ServiceNow ITSM

Get 50% Off Slack Pro

6 days left on this offer

Huddles

Channels

all-new-workspace

itsm

social

Add channels

Direct messages

Selva Arun you

Add coworkers

Apps

Slackbot

ServiceNow BOT

Add apps

itsm

Messages

Approve Decline Today

Only visible to you

Slackbot 9:12 AM

Oh no, something went wrong. Please try that again.

Only visible to you

Slackbot 9:20 AM

Oh no, something went wrong. Please try that again.

Only visible to you

Slackbot 9:31 AM

Oh no, something went wrong. Please try that again.

Only visible to you

Slackbot 9:41 AM

Oh no, something went wrong. Please try that again.

ServiceNow APP 10:03 AM

A New Change Request Needs Approval!

CHG0030010 - Thank you Lord Jesus for helping me !!!!

Approve Decline

Change request CHG0030010 has been approved as per your response.

Only visible to you

Slackbot 10:03 AM

Oh no, something went wrong. Please try that again.

ServiceNow APP 11:32 AM

A New Change Request Needs Approval!

CHG0030011 - Apply the security patch to the servers

Approve Decline

Profile

Selva Arun

+ Add name pronunciation

Active

11:32 AM local time

Set a status View as

Contact information

Email Address

knowservicenow@gmail.com

About me

+ Add Start Date

3. Approvers can click **Approve** or **Decline** directly in Slack.

CHG0030011 | Change Request

dev197363.service-now.com/now/nav/ui/classic/params/target/change_re...

servicenow All Change Request - ... Application scope: Global Update set: Default [Global]

Change Request CHG0030011 Discuss Follow Update

New ✓ Assess Authorize Scheduled Implement Review Closed

Number CHG0030011 Model Normal

Requested by System Administrator Type Normal

Category Other State Authorize

Service On hold ☐

Service offering Conflict status Not Run

Configuration item Conflict last run

Priority 4 - Low * ITSM App-Dev Assignment group

Risk Moderate Assigned to

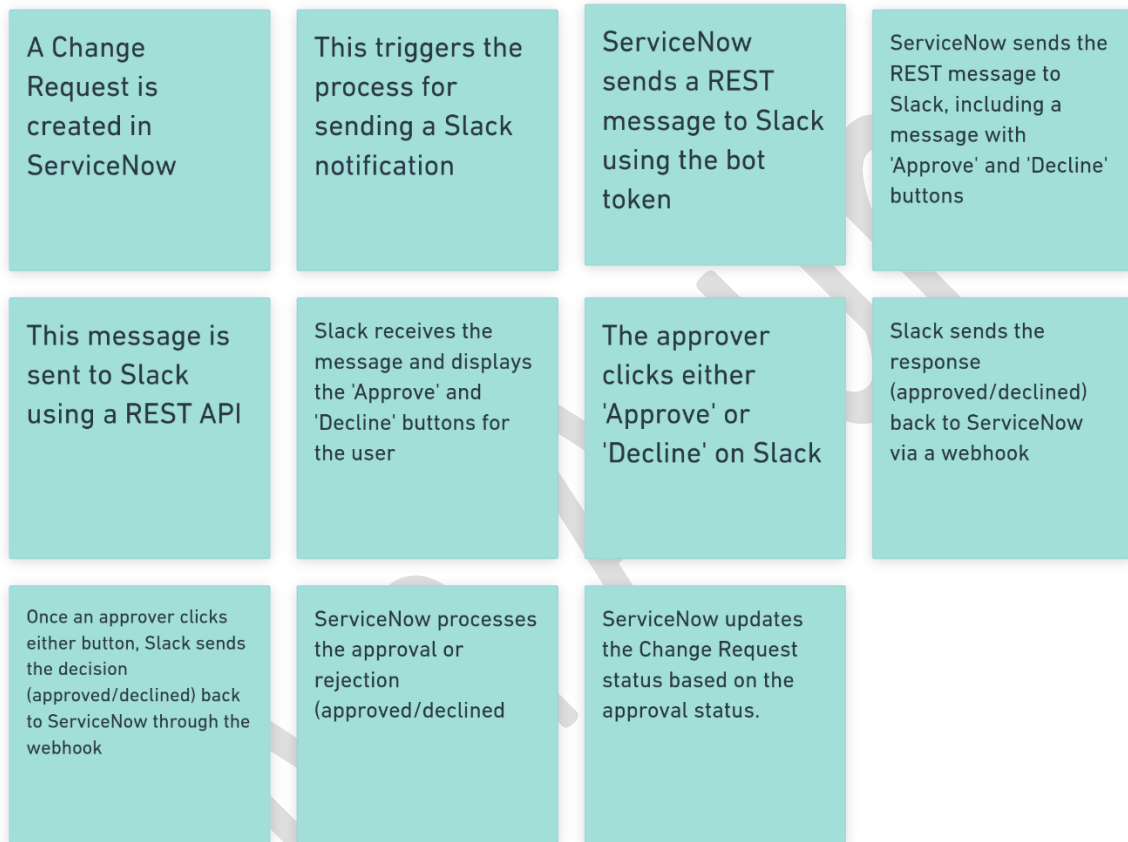
Impact 3 - Low

Short description Apply the security patch to the servers

Description

Planning Schedule Conflicts Notes Closure Information

4. The Change Request status will be automatically updated in ServiceNow, and a **response will be posted back to Slack**, confirming that the Change Request has been updated.



Summary

-  ServiceNow REST Message sends approval requests to Slack.
-  Slack Buttons allow approvers to approve/decline directly.
-  ServiceNow Webhook processes approvals and updates the Change Request status.
-  **Bot Token Scopes:** chat:write, commands, incoming-webhook (for bot interaction).
-  **User Token Scopes:** Ensure the app has the necessary permissions to perform the actions on behalf of users.
-  Ensure that the **Slack App is invited to the #itsm channel** to receive notifications and handle approvals.

 **For a detailed walkthrough, check out our YouTube video:**

[Learn ServiceNow to Slack Integration for Change Request Approvals!](#)

<https://www.servicenow.com/docs/bundle/yokohama-integrate-applications/page/administer/integrationhub/reference/ih-plugins.html>

<https://api.slack.com/>

If you found this helpful, please **like**, **share**, and **subscribe** to our channel. We'd love to hear your feedback in the comments!